

Exit Exam question 2015

1. Which parameter is used to assess and evaluate software architectures?
 - a. Responsiveness of Architectures
 - b. Durability of Architectures
 - c. Architectural quality in attribute
 - d. Number of components in the Architecture's
- 2.
3. What we use to pass the data between activities in Android?
 - a. Broadcast receiver
 - b. PostgreSQL Database
 - c. Intent
 - d. content provider
4. The following UML design shows student class. Assume that the Attributes have declared as private and the method as public. The get method return the value of the attribute and calculateAge() returns the student age. Which of the following would not raise an error when you run this program?

Student
Name: String
Department: String
Phone: String
Age: int
Student (String, String, String, int)
SetDept(String)
GetName(): String
GetDept():String
GetYearOfBirth():int
CalculateAge(): int

- a. Student st1 = newStudent("john", "SE", "0913222222", 20)
 - b. st1 Student = newStudent("john", "CS", "02085332455", 23)
 - c. Student st1 = newStudent("john", "IT", "2000",)
 - d. Student st1 = newStudent("john", "CS", "0913222222", "25")
5. What will be the output of the following code snippet?
itex<script type="text/JavaScript" language = "JavaScript" >

```
var a = "exitexam";  
var result = "a.substring(2,6)";  
document.write(result);
```

- a. texam
- b. xite
- c. itex
- d. xitex

6. Which one of the following is wrong regarding to interface?

- a. Like a class, an interface is reference data type.
- b. Interface abstract method are accessed using interface instances.
- c. Interface includes abstract methods.
- d. One class can implement multiple interfaces.

7.

8. For the schema Hotel (Hotel_id:integer, Hotel_name: String, Sub_city: string) which SQL statement correctly inserts Tuples to Hotel relation. (Assume Hotel_id is primary key).

- a. Insert INTO HOTEL (Hotel_id, Hotel_name, Sub_city)VALUES ('Hilton','Yeka')
- b.
- c.
- d. Insert INTO HOTEL (Hotel_id, Hotel_name, Sub_city)VALUES (1,'Hilton','Yeka')
- e. Insert INTO HOTEL (1,'Hilton','Yeka')
- f. Insert Values INTO HOTEL(Hotel_id, Hotel_name, Sub_city)VALUES (1,'Hilton','Yeka')

9.

10. Which of the following shows the five stages in Tuckman's model of team development, in sequential order?

- a. forming, storming, performing, norming and adjourning
- b. norming ,forming, storming, performing and adjourning
- c. storming, forming, norming, performing and adjourning
- d. forming, storming, norming, performing and adjourning

11. Which Statement is correct about the difference between functional and nonfunctional requirement?

- a. Unlike functional requirement, nonfunctional requirement is stable.

- b. Both are the same most of the time and only vary under certain contexts.
- c. Functional requirements are decided by the customer while nonfunctional requirements are decided by developers
- d. Functional requirements are the service of the system provide while nonfunctional requirements are constraints on the service or functions offered by the system.

12.

13. If the estimate the total discounted benefits for a project is birr 120,000 and total discounted cost is birr 100,000 what is the estimated return on investment (ROI) percentage?

- a. 20%
- b. 12%
- c. 30%
- d. 10%

14. A queue in which the item most recently added is always the first one out refers to:

- a. LIFO queue
- b. FIFO queue
- c. Real time queue
- d. Priority queue

15.

16.

17. A firewall is used in a system connected to a wide area network to

- a. Fire spreading via network through cables
- b. Avoid Spreading of fire in the network
- c. Stop Unauthorized access by hackers
- d. Scanning for viruses in various files

18. The permission –rwxr-r- represented in octal expression will be:

- a. 766
- b. 700
- c. 744
- d. 755

19. Which of the following is the correct order of packet data unit as data moved in the OSI reference model from bottom to top?

- a. Segment, Packet, Frame, Bit
- b. Segment, Frame, Packet, Bit

- c. Bit, Frame, Packet, Segment
- d. Bit, Packet, Frame, Segment

20. Which of the following statement is true regarding triple DES?

- a. 192-bit keys on 64-bit blocks of plain text.
- b. Uses 128-bit blocks of plain text and 92-bit keys and apply DES algorithm thrice.
- c. Work with 192-bit blocks of plain text and applies DES algorithm once.
- d. Functioning on 64-bit blocks of plain text and 56-bit keys by applying DES algorithm for three rounds.

21.

22. An HTTP request message always contains _____?

- a. a status line, a header, and a body
- b. a header only
- c. a header and a body
- d. a request line and a header

23. Which of the following is refers to unauthorized disclosure of information?

- a. authorization
- b. authentication
- c. integrity
- d. confidentiality

24. Anything that can perceive its environment through sensory and act as upon environment through effectors

- a. Agent
- b. Expert System
- c. Intelligence
- d. API

25. Assume that there is a pet family from which Pets like Dog and Cat share behavior from every Pet family has a name. However, the pet families have some specific behaviors that are exhibited by only particular pets. For instance, only Dogs can fetch something while both cats and dogs can speak which of the following object oriented design principle would be suitable for this case?

- a. Information hiding
- b. Encapsulation

c. Polymorphism

d. Inheritance

26. Which of the following statement is not addressing the banker's algorithm?

a. The banker's algorithm considers each request before it occurs.

b. It is an extension of the deadlock detection algorithm.

c. It is modeled on the way a small-town banker might deal with a group of customers to whom he/she has granted lines of credit

d. It is scheduling algorithm.

27. An informed search algorithm that provides exponential time complexity in the worst-case scenarios and guarantees optimality when searching a solution refers to:

a. A* search

b. greedy best-first search

c. uniform cost search

d. Iterative deepening A*search

28. Which principles state that 80% of the problem can be fixed with 20% of the entire effort?

a. Pareto principle

b. Parametric principle

c. Pairwise principle

d. Partition principle

29. Which one of the following is an appropriate sequence that depicts the model development process in data mining?

a. Model training, Model Testing, Model Evaluation, Model Deployment

b. Model Evaluation, Model training, Model Testing, Model Deployment

c. Model training, Model Testing, Model Deployment, Model Evaluation

d. Model training, Model Evaluation, Model Testing, Model Deployment

30. Which method has the same name as that of its class?

a. delete

b. class

c. constructor

d. finalize

31. Within OAuth, what component validates the user's identity?

a. Authorization server

b. resource server

c. browser

d. client

32. The result of the following program after running will be

```
Class PrintResult {  
Public static void main (String arg [ ]) {  
int arr{ }={3,4,5,6,7};  
for(int i=0; i<=arr.length-2; ++i){  
system.out.println(arr[i]+ ",")  
}  
}}
```

- a. 3,4
- b. 3,4,5
- c. 3,4,5,6,7
- d. 3,4,5,6

33. While optimizing our relation, if we found that no multivalued attributes and partial dependencies exist in a relation then the relation is in what normal form?

- a. 1NF
- b. 3NF
- c. 4NF
- d. 2NF

34. A Techniques for generating plans with conditionals and loops are almost identical to those for generating programs from logical specification is called:

- a. Automatic programming
- b. Automatic monitoring
- c. Automatic recursive
- d. Automatic learning

35. Which one of the following is the correct identifier in c++?

- a. 7variable
- b. 7VARIABLE
- c. \$variable
- d. variable_1234

36. Which of the following is true about the abstract class?

- a. cannot be inherited
- b. a mechanism of encapsulation
- c. cannot be instantiated
- d. can contain both abstract and non-abstract methods

37. In software risk management, accepting risk happening but do not do anything is known as
- a. risk transfer
 - b. risk retention**
 - c. risk avoidance
 - d. risk reduction
38. When a process is in a “Blocked” state waiting for some input-output services and after the service is completed it will go to a state known as:
- a. Suspended
 - b. Terminated
 - c. Running
 - d. Ready**
39. “The system architecture should be designed using fine-grain, self-contained components. Producers of data should be separated from consumers and shared data structures should be avoided.” For which requirement this architecture description applies?
- a. Availability
 - b. maintainability**
 - c. performance
 - d. security
40. Which slogan is against agile method philosophy?
- a. Following a plan over responding to changes**
 - b. Individuals and interactions over processes and tools
 - c. Customer collaboration over contract negotiation
 - d. working software over comprehensive documentation
41. APK stands for
- a. Android Package Kit**
 - b. Android Platform Kit
 - c. Android Phone Kit
 - d. Android Page Kit
42. Which statement describes the main goal of software testing as a part of the quality assurance process?
- a. to achieve project timelines and deadlines
 - b. to generate software requirement and design documentation
 - c. to guarantee that the software meets the specified requirement**
 - d. to monitor the entire SDLC

43. The correct structure of the for loop statement is:
- a. for(initialization; condition)
 - b. for(increment/decrement; initialization; condition)
 - c. for(condition, initialization, increment/decrement)
 - d. for(initialization; condition; increment/decrement)
44. Which of the following is not true regarding algorithm development during problem solving?
- a. should be developed by considering the platforms on which it runs
 - b. Is step wise logical description of how to solve the problem
 - c. Is developing considering the details of the programming language
 - d. Is equivalent to the programming language code
45. What would be the most likely consequence if every software stakeholders were rushing to be approved for their requirement without discussing with other stakeholders?
- a. Attains the project schedule
 - b. Results in excess requirements
 - c. leads to conflict requirement
 - d. Requirement become unambiguous
46. Which of the following is true regarding model development in machine learning?
- a. Much focus should be given to test data than training data
 - b. Training with large dataset boosts model performance
 - c. Test set and Training set should have fifty-fifty overlap
 - d. proportion of data set for training should be less than the test set
47. All layout classes are subclasses of which of the following?
- a. android.view.RelativeLayout
 - b. android.view.Layout
 - c. android.view.ViewGroup
 - d. android.view.Widget
48. What will be the output for the following code if you enter 4 and 3 for b and e respectively
- ```
#include<iostream>
Using namespace std;
int main(){
int b, e, r=1;
```

```

count<<"enter b and e";
cin>>b>>e;
for (int i=1; i<=e; i++){
f=r*b;
}
cout<<f;
return 0;
}

```

- a. 81
- b. 12
- c. 64
- d. 256

49. Which of the following is not a valid host address within the network 172.17.128.0/21?

- a. 172.17.128.1/21
- b. 172.17.135.255/21
- c. 172.17.128.255/21
- d. 172.17.135.0/21

50. Which of the following is the correct sequence that will be followed during software evolution process?

- a. change request, release planning, change implementation, impact analysis, system release
- b. impact analysis, change request, change implementation, release planning, system release
- c. change request, impact analysis, change implementation, release planning, system release
- d. change request, impact analysis, release planning, change implementation, system release

51. Identify the lowest layer of android architecture

- a. Linux kernel
- b. Database
- c. Application Framework
- d. Application

52. Which of the following is true regarding the use of switches and hubs for network connectivity in the network?

- a. Switches do not forward broadcasts
  - b. Switches increase the number of collision domains in the network
  - c. Switches take less time to process frames than hubs take
  - d. Hubs can filter frames
53. Select the one that is not correct related to generalization in system modeling.
- a. In generalization, the attribute and operation associated with the highest-level classes are also associated with lowest level classes.
  - b. Common information will be maintained in one place
  - c. Facilitate easy modification of data
  - d. The higher level classes are more specific than the lower-level classes by adding specific attributes and operation
54. A set of process management which is waiting for a resource that is owned by another process refers to:
- a. deadlock
  - b. preemption
  - c. overloading
  - d. queueing
55. A set of processes management which is waiting for a resource that I owned by another process refers to:
- a. Deadlock
  - b. Preemption
  - c. Overloading
  - d. Queuing
56. “Database is a logically coherent collection of data with some inherent meaning “ describes that
- a. All attributes of the entity should be related
  - b. there should be one primary key attribute
  - c. the entity should be physical object
  - d. number of attributes should be limited
57. Which of the following is a resource optimization techniques in which start and finish dates are adjusted based on resource constraints with the goal of balancing the demand resources with the available supply?
- a. Responsibility assignment matrix
  - b. Resource smoothing
  - c. Resource leveling

d. Resource grouping

58. Business people always remodel their business by closely studying their customers purchasing habits over the course of their purchase history. Accordingly, a salesperson at Markato shop noticed that people who always buy shoes will also buy socks with probability of 0.75 and hence decided to put shoes and socks side by side for ease of accessibility. Which big data analytics technique could this salesperson could employ?

- a. Regression Modeling
- b. Clustering algorithm
- c. Predictive modeling
- d. Association rule

59. Given the following layered architecture which ordering relation is correct and didn't encountered any architectural erosion? Assume there are four layers: A,B,C and D

|   |
|---|
| A |
| B |
| C |
| D |

- a. (A,B)
- b. (D,A)
- c. (A,C)
- d. (C,B)

60. Which of the following is among benefits provided with access control lists (ACLs) implementation for software security based application?

- a. ACLs monitor the number of bytes and packets
- b. ACLs provide high network availability
- c. Virus detection
- d. ACLs classify and organize network traffic

61. Default method while Submitting a form is

- a. Post method
- b. Get Method
- c. Put Method
- d. Set Method

62. The process that generates activities such as project schedule, cost estimations and work breakdown structures is described as:

- a. Project Initiating

- b. Project Executing
- c. Project Closing
- d. Project Planning

63. Given the following fragment of code, how many tests are required for 100% decision coverage?

```

 if width > length
 then biggest_dimension = width
 if height > width
 then biggest_dimension = height
 end_if
else biggest_dimension = length
if height > length
then biggest_dimension = height
end_if
end_if

```

- a. 6
- b. 4
- c. 2
- d. 1

64. How can we select an element with a specific class in CSS?

- a. ^
- b. -
- c. #
- d. .

65. Select an activity that suits in the fundamental test process which includes evaluation of the testability of the requirement and system?

- a. Test analysis and design
- b. Test analysis and requirements
- c. Test analysis and implementation
- d. Test analysis and planning

66. Which of the following is an appropriate sequence of database design processes?

- a. Logical design database, Enterprise data modeling, Physical database design, Database implementation
- b. Enterprise data modeling, Logical design database, Database implementation, Physical database design

c. Enterprise data modeling, Logical design database, Physical database design, Database implementation

d. Logical design database, Enterprise data modeling, Physical database design, Database implementation

67. Which of the following is not among the purposes of software testing?

a. Identify shortcomings

b. Improving product acceptance

c. Enhancing reliability

d. Requesting more design and implementation time

68. Which software process model will you use if you want to deliver different functionalities (modules) of the software product that have different priority at different time?

a. Incremental model

b. Spiral model

c. Waterfall model

d. Linear model

69. What is the time complexity for the following code snippets?

```
function someFunction(n)
```

```
 for(var i=0; i<n*10; i++){
```

```
 console.log(n);
```

```
 }}
```

a.  $O(n^2)$

b.  $O(\log n)$

c.  $O(\log^2 n)$

d.  $O(n)$

70. Select the wrong statement regarding traceability in software development projects

a. Traceability shows the dependency among two or more requirements if any

b. Requirements should be linked to the respective stakeholders who generated them

c. Every design element (component) should be linked back to the requirement

d. Traceability traces the overall cost and schedule of every activity in the requirements

71. Which activity comes last in fundamental program development in c++?

- a. Memory allocation for variables
- b. Linking
- c. Compiling program
- d. Writing program

72. Which design pattern is used to restore the state of an object to its previous state?

- a. Visitor pattern
- b. Momento pattern
- c. Observer pattern
- d. Iterator pattern

73. Which of the following describes a set of standards and an associated network protocol that allow establishing a secure channel between a local and a remote computer?

- a. Secure socket layer(SSL)
- b. Secure shell(SSH)
- c. Authentication, Authorization and Accounting(AAA)
- d. Simple network management protocol(SNMP)

74. What components hides the distinctions or boundaries between various micro services from end client application?

- a. A Layered system
- b. API gateway
- c. API proxy
- d. API logging

75. Select a fundamental issue that may not be considered by the software architects during the architectural design process?

- a. Which architectural organization is best for delivering the functional requirements
- b. How to decompose structural components in to sub components
- c. How the system will be distributed across a number of cores or processes
- d. Which architectural patterns or styles to use

76. Which of the following statement is not correct statement regarding of project life cycle?

- a. In agile life cycle the project scope is generally determined early in the project life cycle, but time and cost estimates are routinely modified
- b. In waterfall life cycle, the project scope, time and cost are determined in the early phase of the life cycle.
- c. In agile life cycles, the project scope cannot be outlined and agreed before the start of iteration.
- d. In an incremental life cycle, the deliverable is produced through a series of iterations that successively add functionality within a predetermined time frame.

77. Which of the following is not among the characterizing features of database approach?

- a. Application-data dependency
- b. Sharing of data
- c. Self-describing
- d. Data abstraction

78. What is an activity in android?

- a. android class used to configure the android application
- b. It is an intent
- c. android package file holding all the package used
- d. A single screen in an application with supporting java code

79. Which of the following is not an approach used by IT security specialists to enhance the security level of the network?

- a. Use of Intrusion detection system
- b. Use of Wana cry protocol
- c. Use of physical security
- d. Use of intrusion prevention system

80. As a software tester, when do you implement configuration management procedures?

- a. During test planning
- b. During test closing
- c. During test execution
- d. During test initiation

81. Identify the one that comes first in data mining process

- a. Data cleaning
- b. Data integration
- c. Business understanding
- d. Data selection

82. A TCP/IP layer which is responsible for addressing, path selection and routing refers:

- a. Application layer
- b. Internet layer
- c. Transport layer
- d. Network Access layer

83. Which of the following analysis mechanism defines the input for which the algorithm takes the least time (fastest time to complete)?

- a. Worst case
- b. Best case
- c. Standard case
- d. Average case

84. Which learning algorithm is usually applied to data that does not contain any label information, so we do not know what the right output should be?

- a. Transfer learning
- b. Supervised learning
- c. Unsupervised learning
- d. Reinforcement learning

85. Depending on the organization of the elements which of the following approach is a non-linear data structure?

- a. Queue
- b. Stack
- c. Tree
- d. Linked list

86. Which one of the following can be among possible source of unstructured data?

- a. Apple employee database
- b. RDBMS systems
- c. Registrar student database
- d. Google search engine

87. Select the one that is not related with the purpose of properly documenting software architecture
- a. For Stakeholder communication
  - b. To easily figure out the source code flow
  - c. For critical system design decisions
  - d. To know how the system is organized and interoperate
88. One of the following statement is correct about Artificial Intelligence (AI)?
- a. It refers to the development of machine that can only perform physical tasks.
  - b. It refers to the development of machine that can only perform simple, repetitive tasks.
  - c. It refers to the development of machine that can perform tasks only in a controlled laboratory setting.
  - d. It refers to the development of machine that can perform tasks that typically require human intelligence.
89. Which of the following is a process of selecting program paths in such a manner that certain branches (i.e, outgoing edges of nodes) of a control flow graph are covered by the execution of those paths?
- a. Branch coverage
  - b. Branch expansion
  - c. Branch control
  - d. Branch boundary
90. Give a situation two hosts attempt on half-duplex Ethernet LAN to send data concurrently, resulting in a collision and subsequent to this collision, what will the hosts do?
- a. The jam signal indicates that the collision has been cleared
  - b. An electrical pulse indicates that the collision has cleared
  - c. The hosts will attempt to resume transmission after a time delay has expired
  - d. The router on the segment will signal that the collision has cleared

91. What is the time complexity of below function?

```
Public void function (int n){
 for(int i=0; i<n; i++)
 for(int j=0; j<i*i; j++)
 if(j%i==0){
 for(int k=0; k<j; k++)
 print("*");
 }
 }
 }
```

- a.  $O(n^2)$
- b.  $O(n \log n)$
- c.  $O(n^7)$
- d.  $O(n^5)$

92. Identify the design principle that does not apply to software system

- a. design should be traceable to the analysis model
- b. design should be structured to accommodate change
- c. design should exhibit uniformity and integration
- d. design should be reinventing the wheel from scratch

93. You are required to write a program that iteratively takes 50 numbers from user input and adds them up only if the numbers are positive and skips if not. Which control structure best fits this scenario?

- a. break
- b. continue
- c. for
- d. jump

94. Which of the following UML elements is wrongly applied?

- a. Class diagram shows the object classes in a system and their relation
- b. Activity diagram shows how the system reacts to internal and external events
- c. Use case diagram represents an interaction with a system
- d. Sequence diagram shows the sequence of interactions required to complete some operation

95. A memory management policy that that decisions must be made as to which page or pages are to be supplanted when a memory is full is known as:

- a. Cleaning policy

- b. Load policy
  - c. Replacement policy
  - d. Fetch policy
96. Which software model would you apply for business operating in a changing environment where the initial requirements in inevitably change due to some internal and external factors?
- a. Agile development method
  - b. Spiral model
  - c. Water fall model
  - d. Code and run
97. Assume that you are assigned to design a dormitory management database. A dormitory is located on some building and has attributes specified as DORM (DormId, floornumber) and a DormId is primary key based on university's rule, six students can be assigned to one dorm. Students can be tracked with attributes STUDENT (IdNo, Name, Departement) and IdNo is a primary key. Based on this scenario description, which of the following SQL statement is correct to establish relationship between DORM and STUDENT assuming DorId is foreign key?
- a. FOREIGN KEY (DorID) REFERENCES DORM(DormID)
  - b. STUDENT table is a reference relation
  - c. FOREIGN KEY (DorID) REFERENCED by DORM(DormID)
  - d. DORM table is a referencing relation
98. Which one of the following statements is wrong regarding try...catch construct?
- a. When exception occurs, the execution of try block is interrupted
  - b. The execution control is moved to catch block after exception occur in try block.
  - c. The catch block is used to handle exceptions that occur in try block.
  - d. The catch block is executed once regardless of exceptions in try block
99. Which of the following is not the function of Cross-over UTP cable?
- a. To connect switch to switch
  - b. To connect pc to pc
  - c. To connect router to pc
  - d. To connect switch to pc
100. Who is responsible for documenting faults found during the software development process?

- a. Tester
- b. Developer
- c. Scrum master
- d. Requirement engineer

## MODEL

1. In a quad-tree each non-leaf node has exactly \_\_\_\_\_ children.
  - a. 1
  - b. 3
  - c. 2
  - d. 4
2. Which of the following keyword is used to define a variable in javascript?
  - a. const
  - b. int
  - c. let
  - d. val
3. The incredible change in business and society as emerging economies develop and new technologies become available mainly.
  - a. Makes obsolete the already existing software systems.
  - b. Lead to demand for new software architecture.
  - c. require new testing on the existing software systems
  - d. Lead to privacy and security breaches
4. Which one of the following is not correct?

- a. HTML form elements are used for taking user input.
  - b. HTML form elements can be of different types.
  - c. HTML form elements are used for displaying outputs.
  - d. HTML form elements are defined inside < form > tag.
5. A process scheduling approach on which decision as to which available process will be executed by the processor refers to.
- a. I/O scheduling
  - b. Medium-term scheduling
  - c. Long-term scheduling
  - d. Short-term scheduling

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1. A process scheduling approach on which decision as to which available process will be executed by

the Processor refers to:

- a. Long-term scheduling
- b. I/O scheduling
- c. Short-term scheduling
- d. Medium-term scheduling

c. Short-term scheduling

2. Which of the following analysis mechanism defines the input for which the algorithm takes the least time (fastest time to complete)?

- a. Best case
- b. Average case
- c. Standard Case
- d. Worst case

a. Best case

3. Which of the following UML diagram is not used to represent dynamic behavior of an object during analysis?

- a. Instance diagram
- b. Sequence diagram.
- c. Collaboration diagram
- d. Activity diagram

a. Instance diagram

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4. Assume there is Class A which is extended by Class B and Class C  
Class A is declared as abstract,

and also contains an abstract method called welcome; this method takes no parameters, and returns a

String. Which of the following definitions of Class B will NOT cause a compiler error based on the above scenario?

a. public class Class B extends Class A

```
{
public String whatObjectAmI()
{
return "I am a class B object";
}
}
```

b. public class Class B extends Class A

```
{
public void welcome(String str)
{
}
}
```

c. public class Class B extends Class A

```
{
public String whatObjectAmI()
{
return "I am a class 8 object";
}
```

```
}
```

d. public class Class B extends Class A

```
{
```

```
public void welcome(String str)
```

```
{
```

```
}
```

```
}
```

b. public class Class B extends Class A { public void welcome(String str) { } }

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5. Which of the following is not type of Ensemble learning?

a. Lasso

b. Boosting

c. Bagging

d. Random forest

a. **Lasso**

6. A technique for generating plans with conditionals and loops are almost identical to those for generating programs from logical specifications is called \_\_\_\_.

a. Automatic programming

- b. Automatic learning
- c. Automatic monitoring
- d. Automatic recursive

**a. Automatic programming**

7. The CSS property used to specify the transparency of an element is?

- a. Transparency
- b. Opacity
- c. Background
- d. Alpha

**b. Opacity**

8. Which of the following describes the functionalities of application layer?

- a. It deals encapsulation of frames
- b. It deals encapsulation of packets
- c. It deals mechanical, electrical and physical connectivity
- d. It is process to process interaction

**d. It is process to process interaction**

9. Which function is used to serialize an object into a JSON string in Javascript?

- a. `parseJson()`
- b. `toJson()`
- c. `stringify()`
- d. `parse()`

c. `stringify()`

10. A TCP/IP port number used by SSH is?

- a. 20
- b. 22
- c. 23
- d. 21

b. 22

11. Given the following fragment of code, how many tests are required for 100% decision coverage?

if width > length.

then biggest\_dimension = width

if height > width

then biggest\_dimension = height

end\_if

```
else biggest_dimension = length
if height > length
then biggest_dimension = height
end_if
end_if
```

- a. 2
- b. 6
- c. 4
- d. 3

c. 4

12. Which of the following is true regarding the referential triggered action ON UPDATE CASCADE

- a. sets the referencing primary key to NULL
- b. change the value of the referencing foreign key attribute to the updated primary key value for all the referencing tuples
- c. sets the referencing primary key of the base table to default value
- d. deletes all tuples of the referencing table

b. change the value of the referencing foreign key attribute to the updated primary key value for all the referencing tuples

13. Which file system can be used to change certain kernel parameters at runtime using sysctl command?

- a. Ext4
- b. Sysfs
- c. Procfs

**c. Procfs**

- d. Ext3

14. A software tool, such as a design editor or a program debugger, that developers and other software

stakeholders use to support an activity in the software development process is

- a. CASE tools
- b. CMMI
- c. CMM
- d. BPMN

**a. CASE tools**

15. Which of the following statements designates part of the process of how a switch resolves to forward a frame destined for a recognized unicast MAC address?

- a. It matches the unicast destination address to the bridging, or MAC address, table.
- b. It matches the unicast source address to the bridging, or or MAC address, table.
- c. It matches the frame's incoming interface to the source MAC entry in the MAC address table.
- d. It matches the destination IP address to the destination MAC address.

a. It matches the unicast destination address to the bridging, or MAC address, table.

16. Which of the following is a unique IPV4 address?

- a. 172.17.17.17/26
- b. 172.16.30.0/16
- c. 172.16.25.250/16
- d. 172.43.17.1/16

a. 172.17.17.17/26

17. Which one of the following is not correct?

- a. HTML form element can be of different types
- b. HTML form elements are used for taking user input
- c. HTML form elements are used for displaying outputs
- d. HTML form elements are defined inside <form> tag

c. HTML form elements are used for displaying outputs

18. One of the following statements addresses the characteristics of functional organization structure?

a. Low to moderate project manager authority, low to moderate resource availability, and a full-time

project manager's role

b. Little or no project manager authority, little or no resource availability, and the functional manager control the project budget =>  
weak matrix

c. High to almost total project manager authority, high to almost total resource availability, and

full-time project management administrative staff

d. Limited project manager authority, limited resource availability, and a part-time project manager's

role

d. Limited project manager authority, limited resource availability, and a part-time project manager's role

19. What will be the output of the following program

```
class increment {
```

```
public static void main(String args[])
```

```
{
```

```
int a=4;
System.out.print(++a * 5);
}
}
```

- a. 5
- b. 20
- c. 24
- d. 25

d. 25

20. In a quad-tree each non-leaf node has exactly \_\_\_\_ children.

- a. 2
- b. 1
- c. 4
- d. 3

c. 4

21. Which of the following mostly used by software project quality evaluators to using the approach of plan, check and act?

- a. IEEE
- b. Capability maturity model
- c. Six sigma
- d. ISO model

c. Six sigma

22. The result of the following program after running will be:

```
class PrintResult
{
public static void main(String args[])
int arr[] = {3, 4, 5, 6, 7};
for (int i = 0;i < arr.length - 2; ++i)
System.out.println(arr[i] + " ");
}
}
}
```

- a. 3 4 5
- b. 3 4 5 6
- c. 3 4
- d. 3 4 5 6 7

a. 3 4 5

23. An Artificial Intelligence system is composed of:

- a. Agent and Environment
- b. Data and Information
- c. Device and Network
- d. Technology and Evolution

a. Agent and Environment

24. Which of the following is true regarding integration testing?

- a. It uses test data that simulates ideal scenarios.
- b. It emphasizes on testing individual software components in isolation.
- c. It proves the functioning of different software components when integrated together.
- d. It is only performed after system testing is completed.

c. It proves the functioning of different software components when integrated together.

25. Which of the following describes the leading agile development method?

- a. Sprint
- b. Six sigma
- c. Extreme programming
- d. Scrum

d. **Scrum**

26. Which access modifier is suitable if you want your instance variables to be accessible from both the methods of the class and the descendant classes?

- a. Private
- b. Inherited
- c. Public
- d. Protected

d. **Protected**

27. Which of the following is not a data link layer technology?

- a. OSPF
- b. VLAN
- c. PPP

d. Frame-relay

a. OSPF

28. The purpose of <iostream> header file in c++ program is:

a. Makes the code secure

b. to make the code readable

c. to provide input/output functionality

d. to create file streams

c. to provide input/output functionality

29. What of the following describes informed search algorithms?

a. They are optimal

b. They are admissible

c. They are consistent

d. They are complete

b. They are admissible

30. Which one is not an architectural style?

a. MVC

- b. Client server
- c. UML
- d. OSI layer

c. UML

32. Which of the following specifies divergence between Dijkstra's Algorithm (DA) and Uniform Cost

Search (UCS) algorithms?

- a. DA is optimal, but not UCS
  - b. UCS finds the optimal solution while DA
  - c. DA first collects them into a Queue but UCS discover them as they come
  - d. DA discovers nodes as they come while UCS first collects them in a Queue
- d. DA discovers nodes as they come while UCS first collects them in a Queue

33. In software testing, a term which relates the genuine imperfect segments to the absolute number of segments anticipated as faulty known as:

- a. F-measure
- b. Recall

- c. Accuracy
- d. Precision

d. Precision

34. Which of the following is not true about binary tree?

- a. Empty tree is also a valid binary tree
- b. We can visualize a binary tree as consisting of a root and two disjoint binary trees
- c. A tree is called binary tree if each node has zero child, one child or two children
- d. A tree is called binary tree if and only if each node has zero child

d. A tree is called binary tree if and only if each node has zero child

35. What will happen if there is no break statement next to case labels in switch statement?

- a. All non-matching case labels will be executed
- b. The program generates exception
- c. The default block will be executed
- d. The program will halt

a. All non-matching case labels will be executed

36. Security estimation is an essential component of \_\_\_\_\_ process.

- a. Software design
- b. Software development
- c. SRS design
- d. Risk management

a. Software design

37. Which of the following is one of the indirect applications of queues?

- a. Multiprogramming
- b. Operating systems schedule jobs (with equal priority) in the order of arrival (e.g., a print queue).
- c. Auxiliary data structure for algorithms
- d. Simulation of real-world queues such as lines at a ticket counter or any other first-come first-served scenario requires a queue.

b. Operating systems schedule jobs (with equal priority) in the order of arrival (e.g., a print queue).

38. Which development model helps to cope up with software requirement change to reduce the cost of rework?

- a. staged delivery
- b. prototyping
- c. Spiral model

d. Software Version control

c. Spiral model

31. As a software tester, which testing principle can save time and cost of testing?

- a. Exhaustive testing is not possible
- b. Testing early development process saves time and cost
- c. Prioritize testing based on risk and criticality
- d. Testing the entire development process affect time and cost

c. Prioritize testing based on risk and criticality

39. A temporary endeavor undertaken to create a unique product, service, or result is known as:

- a. Program
- b. Portfolio
- c. Project
- d. Process

c. Project

40. Select the one different from others

a. System design

- b. Requirement validation
- c. Requirement elicitation
- d. Requirement analysis

a. System design

41. One of the following is model optimization technique:

- a. Gradient descent
- b. Bagging
- c. Lasso
- d. Gradient boosting

c. Lasso

42. The keyword used in the SELECT clause to eliminate duplicate tuples from the result of an SQL query is

- a. UNIQUE
- b. DISTINCT
- c. Group By
- d. PRIMARY

b. DISTINCT

43. The ultimate purpose of inheritance in OOP is:

- a. Data protection
- b. Code reusability
- c. Resources saving
- d. Providing multiple forms for methods

**b. Code reusability**

44. The base class from which all classes in Java are inherited?

- a. java.lang.class
- b. java.class.inherited
- c. java.lang.Object
- d. java.class.object

**c. java.lang.Object**

45. Which one is not among the layouts available in android?

- a. Card Layout
- b. Linear Layout
- c. Relative Layout
- d. Frame Layout

**a. Card Layout**

46. Which CSS element is used to make rounded borders around elements?

- a. border-circle
- b. border-rounded
- c. border-radius
- d. border-corner

c. border-radius

47. A technique that allows large numbers of ideas to be classified into groups for review and analysis

- a. Affinity Diagrams
- b. Activity-on-Node
- c. Activity List
- d. Adaptive Life Cycle

a. Affinity Diagrams

48. Which of the following refers to unauthorized disclosure of information?

- a. Integrity
- b. Confidentiality
- c. Authentication

d. Authorization

b. Confidentiality

49. Which of the following is not the contribution of the big data analysis

a. To introduce new culture of data usage

b. To gain competitive advantage

c. To cope up with volatile market

d. To satisfy customer need

d. To satisfy customer need

50. The correct structure of the for loop statement is:

a. for(initialization, condition, increment/decrement

b. for(initialization; condition)

c. for(increment/decrement; initialization; condition)

d. for(initialization; condition; increment/decrement)

d. for(initialization; condition; increment/decrement)

51. Introducing new requirement at the later phase of the project is not recommended due to?

a. Cost of change

- b. coding complexity
- c. analysis complexity
- d. requirement complexity

52. Being a software tester you want to verify the integration of the message between different processes communication using \_\_\_\_\_ .

- a. Network authorization
- b. Decryption algorithm
- c. Message Digest
- d. Digital Signature

53. Fundamental program structure consists of the following core components except

- a. back end component
- b. Variable definition
- c. main0 method
- d. Input/output features

54. Assume you have the following hash table, implemented using linear probing. The hash function you are using is the identity function,  $f(x) = x$ . In which order could the elements have been added to the hash table? There are several correct answers, and you should give all of them. Assume that the hash table has never been resized, and no elements have been deleted yet.

- a. 12, 14, 3, 9, 4, 18, 21
- b. 9, 14, 4, 18, 12, 3, 21
- c. 12, 9, 18, 3, 14, 21, 4
- d. 12, 3, 14, 18, 4, 9, 21

55. If your software system can continue in spite of unexpected problems from internal and external environment, we say your system has good:

a. Integrity

b. Robustness

c. Maintainability

d. Reliability

56. What does velocity in big data refer to?

a. Speed of storage consumption

b. Speed of data ingestion

c. Speed of input data generation

d. Speed of data processing and storing

57. Which of the following android component displays the part of an activity on screen?

a. Manifest

b. Intent

c. View

d. Fragment

58. Assume that you are assigned to design a dormitory management database. A dormitory is located on some building and has attributes specified as DORM (DormID, floornumber) and DormID is a primary key. Based on university's rule, six students can be assigned to one dorm. Students can be tracked with attributes STUDENT (IdNo, Name, Department) and IdNo is primary key. Referring to this scenario, if there is a rule that says every DORM must have at least one student, which of the following is true?

- a. STUDENT has a total participation
- b. STUDENT has existence dependency
- c. DORM has partial participation
- d. DORM has existence dependency

59. A program that initializes all aspects of the system, from CPU registers to device controllers and the contents of main memory, and then starts the operating system known as:

- a. Bootstrap
- b. Cache memory
- c. Bootloader
- d. BIOS

60. Method overriding is a characteristic of which OOP principle?

- a. Abstraction
- b. Inheritance
- c. Encapsulation
- d. Polymorphism

61. A tool that is used to check the quality of software to ensure that it meets some minimum coding standards.

- a. Code Auditor
- b. Documenters
- c. Test Data Generator
- d. Interactive Debuggers

62. A system software that resides in main memory is

- a. Compiler

**b. Loader**

c. Executor

d. Linker

63. Which object-oriented principle allows the creation of several methods with the same name that differ from each other by the type of input and output?

a. Overriding

b. Inheritance

c. Encapsulation

**d. Polymorphism**

64. Which one is not the criteria for selecting best programming language for your problem:

a. Modularity

**b. Platform dependency**

c. Portability

d. Code efficiency

65. Which of the following virtual machines is used by the Android operating system?

a. CLR

b. Docker

**c. Dalvik virtual machine**

d. JVM

66. Which of the following is the first callback method that is invoked by the system during an activity life-cycle?

a. onRestart() method

**b. onCreate() method**

c. onClick() method

d. onStart() method

67. APK stands for:

a. Android Phone Kit

**b. Android Package Kit**

c. Android Page Kit

d. Android Platform Kit

68. Which principle states that 80% of the problems can be fixed with 20% of the entire effort?

a. Pairwise principle

b. Partition principle

**c. Pareto principle**

d. Parametric principle

69. Select the one that is not correct related to generalization in system modeling.

a. In generalization, the attributes and operations associated with higher-level classes are also

associated with the lower-level classes

**b. The higher-level classes are more specific than the lower-level classes by adding specific**

attributes and operations

c. facilitate easy modification of data

d. common information will be maintained in one place

70. Action attribute in HTML forms specifies

- a. The auto completion of form
- b. Which HTTP method is used
- c. Which action is going on

**d. Where to submit form**

71. Which of the following indicates the assurance that a product, service, or result meets the needs of the customer and other identified stakeholders?

- a. Variance
- b. SWOT Analysis
- c. **Validation**
- d. Verification

72. In a relation design, primary key is a critical constraint that should be considered. It should be defined:

- a. Non editable
- b. **NOT NULL**
- c. Within limited value
- d. NULL

73. Adding additional header information as data moved from the application layer to down refers to:

- a. De-capsulation
- b. Routing
- c. Switching
- d. **Encapsulation**

74. The incredible change in business and society as emerging economies develop and new technologies become available mainly:

a. Makes obsolete the already existing software systems

**b. Leads to demand for new software architecture**

c. Requires new testing on the existing software systems

d. Leads to Privacy and security breaches

75. A software work product which is associated with the test execution process group is known as:

**a. Test Reports**

b. Requirements

c. Test Cases

d. Code

76. One of the following statements is inappropriate about Artificial Intelligence?

a. An agent perceives its environment through effectors.

**b. It is a preprogrammed system,**

c. AI is a multidisciplinary field of study.

d. Ability to learn from the environment

77. Which of the following concept describes bootstrapping?

a. A language interpret other language program

**b. A language compile itself**

c. A language compiling other language program

d. All of the above

78. Identify the one that does not affect architectural decision

a. Architectural style to be used

**b. Type of the application to be developed**

c. The way the data is stored

d. Expected performance of the system

79. You fit a model too closely to the particularities of the training set and obtain a model that works well on the training set but works poorly on a new data. This scenario is known as:

**a. over fitting**

b. sweet spot

c. under performing

d. under fitting

80. For the software architecture to be easily understood by other design experts for later maintenance and scalability, there should be

a. Architectural optimization techniques

**b. Architectural notation and semantic description**

c. Architectural refactoring

d. Architectural reverse engineering

81. A modification of the free-list approach is to store the addresses of n free blocks in the first free block.

a. Grouping

b. Counting

**c. Linked-list**

d. Deadlock

82. If performance is the critical requirement of your architecture, which architectural approach you apply while architecting your system?

- a. Increase number of component communication
- b. Increase number of components

**c. Localize critical operations**

- d. Distribute operations across network

83. What is the time complexity of below function?

```
Public void function (int n) {
```

```
for (int i = 0; i<n; i++)
```

```
for(int j=i; j<i*i; j++)
```

```
if(j % i == 0) {
```

```
for (int k = 0;k < j; k++)
```

```
print(" * ");
```

```
}
```

```
}
```

- a.  $O(n^7)$

**b.  $O(n^2)$**

- c.  $O(n^5)$

- d.  $O(n \log n)$

84. Database approach supports different user groups who might have different interest in different parts of the database. This characteristic of data base describes:

- a. Parallel transaction
- b. Multiple users

### **c. Multiple views**

d. Concurrent transaction

85. To reduce the dispute that might arise between the customer and contractor, each requirement should be documented and later checked during product delivery whether each documented requirements are fully implemented or not. What type of requirement validation is this?

a. Realism check

b. Verifiability

### **c. Completeness check**

d. Validity check

86. Select the correct statement that describes local search algorithms (LSA):

**a. The quality of the solution found by LSA depends heavily on the starting point and the choice of the neighborhood function**

b. LSA are typically used for solving convex optimization problems.

c. The time complexity of LSA is independent of the size of the problem.

d. LSA always guarantee to find the globally optimal solution.

87. One of the following is not purpose of software testing?

a. Improving product acceptance

b. Enhancing reliability

c. Identifying shortcomings

### **d. Requesting more design and implementation time**

88. Which of the following statements is false when a packet is being compared to an-access list?

a. It's always compared with each line of the access list in sequential order.

**b. Until all lines have been analyzed, the comparison is not over.**

c. There is an implicit "deny" at the end of each access list

d. Once the packet matches the condition on a line of the access list, the packet is

89. You have received an e-mail which is unsolicited and commercially known as?

a. Wana Cry

**b. Spam**

c. Trash

d. Adware

90. For which of the following Android is mainly developed?

a. Servers

**b. Mobile devices**

c. Laptops

d. Desktops

91. Which activity is not related to data preprocessing?

a. Data transformation

b. Data cleaning

**c. Data optimization**

d. Data reduction

92. An input stream class in java that is used to read data from file is

a. FileOutputStream

b. PipedInputStream

c. **FileInputStream**

d. RandomAccessFile

93. A data link sub layer which is responsible for identifying Network layer protocols and then encapsulating them is known as:

a. Point to point Encapsulation

b. Media Access Control

c. Frame relay

**d. Logical Link control**

94. Assume you have an interface called Payable as follows

```
public interface Payable{
 double getPaymentAmmount();
```

If you have another class Invoice that implements this interface and  
getQuantity() and

getPricePerItem() methods that returns quantity and price per item,  
which of the following is the

correct implementation?

a. public class Invoice extends Payable(  
public double getPaymentAmmount0l

```
return getOuantity() * getPricePerItemO;
}
```

b. public class Invoice implements Payable{  
public getPaymentAmmount(){

```
return getQuantity() * getPricePerItem();
}
```

**c. public class Invoice implements Payable{**

```
public double getPaymentAmmount(){
 return getQuantity() * getPricePerItem();
```

```
d. public abstract class Invoice implements Payable{
 public double getPaymentAmmountO(
 return getQuantity() * getPricePerItem0:
 }
}
```

95. Refer to the following schemas. Attributes underlined are primary keys

HOTEL (Hotel\_id integer, Hotel\_Name: string, Sub city:: string)

ROOM (Room\_id: integer, Hotel\_id: integer, Price: integer)

BOOKING (Hotel id: integer, Room\_id: integer, Date\_from: date, Date\_to: date)

GUEST (Guest\_id: integer, Name: string, Country: string)

Considering the above schema, which of the following SOL statement retrieves Hotel names with their

respective Room\_id

a. SELECT Hotel\_Name, Room\_Id FROM HOTEL, ROOM WHERE ROOM.Hotel\_id= HOTEL.

Hotel\_id

b. SELECT Hotel\_Name, Room\_Id FROM HOTEL, ROOM WHERE ROOM.Room\_id= HOTEL.

Hotel\_id

c. SELECT Hotel\_Name, Room\_Id FROM HOTEL, ROOM WHERE ROOM. Room\_id= HOTEL.

Hotel\_id and ROOM. Hotel\_id=HOTEL. Hotel\_id

d. SELECT Hotel\_Name, Room\_Id FROM HOTEL, ROOM

96. Which one is different from others?

a. Requirement elicitation

b. Requirements validation

c. Requirements analysis

**d. Requirements status tracking**

97. Which of the following keywords is used to define a variable in Javascript?

a. const

b. int

c. val

**d. let**

98. From the following, which one is not among the challenges that are faced by requirement engineers during requirement elicitation?

a. Conflicting requirements

**b. Obsolete requirement**

c. requirement change

d. Negative stakeholders

99. When running your program, you did not see any error. However, the final output you got is not correct. What type of error is this?

a. Compiling error

b. Syntax error

**c. Logical error**

d. Run time error

100. Which of the following is against the professional ethics of software engineering?

**a. Accepting any works regardless of your competence**

b. Responsibly remain up-to-date in your profession

c. Respect the confidentiality of your employers or clients

d. Use your technical skills responsibly

51. a. Cost of change

52. c. Message Digest

53. a. back end component

54. All of the given orders are possible:

- a. 12, 14, 3, 9, 4, 18, 21

- b. 9, 14, 4, 18, 12, 3, 21

- c. 12, 9, 18, 3, 14, 21, 4

- d. 12, 3, 14, 18, 4, 9, 21

55. b. Robustness

56. d. Speed of data processing and storing

57. d. Fragment

58. b. STUDENT has existence dependency

59. a. Bootstrap

- 60. b. Inheritance
- 61. a. Code Auditor
- 62. c. Executor
- 63. d. Polymorphism
- 64. b. Platform dependency
- 65. c. Dalvik virtual machine
- 66. b. onCreate() method
- 67. b. Android Package Kit
- 68. c. Pareto principle
- 69. b. The higher-level classes are more specific than the lower-level classes by adding specific attributes and operations
- 70. d. Where to submit form
- 71. c. Validation
- 72. b. NOT NULL
- 73. a. De-capsulation
- 74. b. Leads to demand for new software architecture
- 75. a. Test Reports
- 76. b. It is a preprogrammed system.
- 77. b. A language compiles itself.
- 78. b. Type of the application to be developed.
- 79. a. Overfitting.
- 80. b. Architectural notation and semantic description.
- 81. c. Linked-list.

82. c. Localize critical operations.
83. c.  $O(n^5)$ .
84. c. Multiple views.
85. c. Completeness check.
86. a. The quality of the solution found by LSA depends heavily on the starting point and the choice of the neighborhood function.
87. d. Requesting more design and implementation time.
88. a. It's always compared with each line of the access list in sequential order.
89. b. Spam.
90. b. Mobile devices.
91. c. Data optimization.
92. c. FileInputStream.
93. d. Logical Link Control.
94. c. public class Invoice implements Payable{  
    public double getPaymentAmount(){  
        return getQuantity() \* getPricePerItem();  
    }  
}
95. c. SELECT Hotel\_Name, Room\_Id FROM HOTEL, ROOM  
WHERE ROOM.Room\_id=HOTEL.Hotel\_id.
96. d. Requirements status tracking.
97. d. let.
98. b. Obsolete requirement.

99. c. Logical error.

100. a. Accepting any works regardless of your competence.